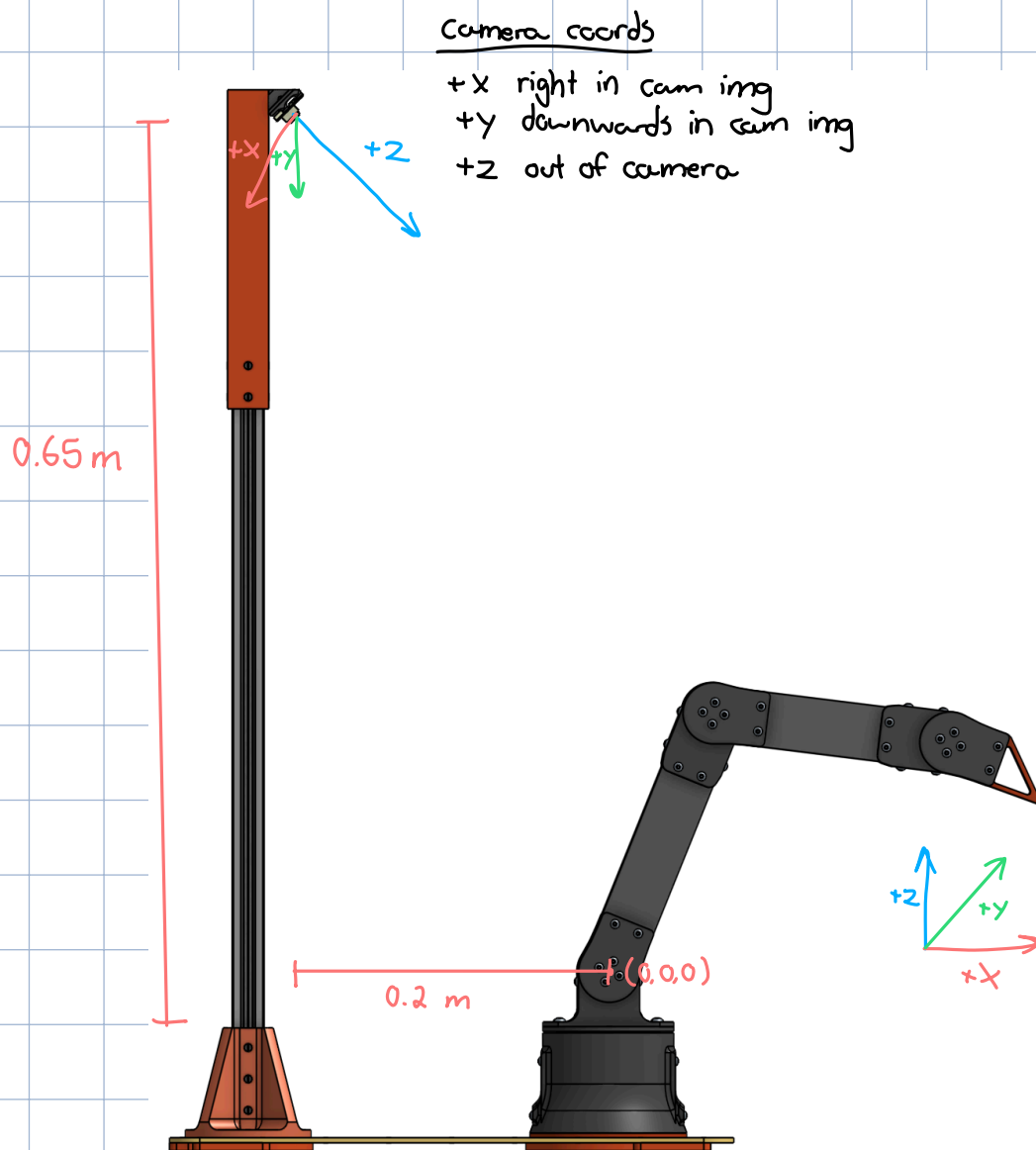




Separate into translation and rotation

$$P_{base} = R_matrix \times P_{cam} + t_cam_base$$

R_matrix is 3×3 rotation matrix

t_cam_base 3×1 translation matrix

Translation:

relative to robot origin:

20 cm behind

65 cm up

5 cm left ($\frac{baseline}{2}$ offset for left cam)

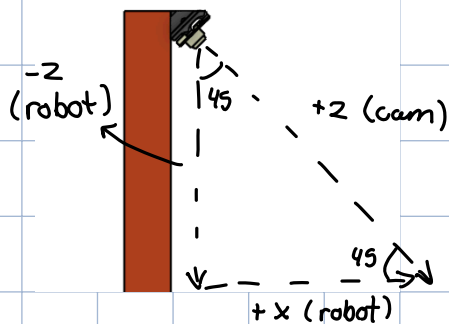
$[-0.2, 0.05, 0.65]$

Rotation

Final matrix is $[\text{cam} + x, \text{cam} + y, \text{cam} + z]$

Cam + z

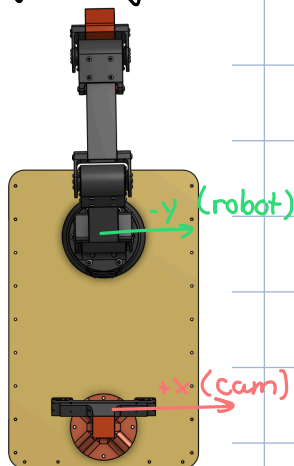
this is forwards and down



$$[\cos 45, 0, -\sin 45]$$

cam + x

right in img

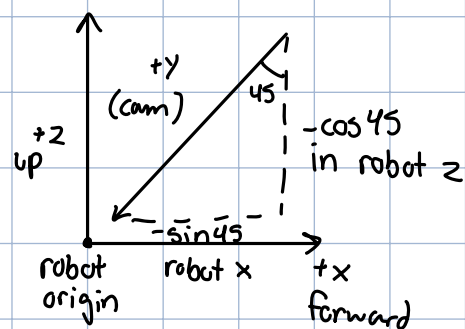


$$[0, -1, 0]$$

Cam + y

downwards in img, so

backwards and down



$$[-\sin 45, 0, -\cos 45]$$

$$\begin{matrix} & x & y & z \\ \begin{bmatrix} 0 & -\sin 45 & \cos 45 \\ -1 & 0 & 0 \\ 0 & -\cos 45 & -\sin 45 \end{bmatrix} \end{matrix}$$

As 4x4 homogenous transform

$$\begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -\sin 45 & -\cos 45 & -0.2 \\ -1 & 0 & 0 & 0.05 \\ 0 & -\cos 45 & -\sin 45 & 0.65 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$